Chapter 7

EIGRP

EIGRP stands for Enhanced Interior Gateway Routing Protocol. It is a contemporary, widely implemented routing protocol with full support for classlessly addressed networks. The basic characteristics of EIGRP are given in Table I.

TABLE I Routing characteristics of EIGRP

Metric	Bandwidth + delay
Interior or Exterior?	Interior
Distance vector or Link-state?	Hybrid
Singlepath or Multipath?	Multipath
Broadcast or Multicast?	Multicast and unicast
Flat or Hierarchical?	Flat
Classful or Classless?	Classless

The advantages of EIGRP include

- Classless protocol—full support for classless addressing
- Scalable protocol—can be used on large networks
- Supports multiple networking protocols—single routing protocol solution for IP, IPX, and Appletalk
- Optimized for minimum bandwidth use
- Can support many hops (up to 255)
- Fast convergence
- Guaranteed loop-free routes

The disadvantages of this protocols include

- Proprietary to Cisco IOS—must employ all Cisco routers to avoid having to run multiple interior routing protocols
- Flat protocol—cannot be configured into areas as OSPF can

Differences between EIGRP and RIP are given in Table II.

TABLE II RIP-versus-EIGRP

	RIP	EIGRP
1	It support maximum 15 routers in the network. 16 router is unreachable.	It supports maximum 255 routers in the network. However, the default is 100 routers. (highly scalable)
2	Slow convergence	Fast convergence due to feasible successor
3	In RIP routing protocol, we cannot create a separate administrative boundary in the network.	In EIGRP routing protocol we can create a separate administrative boundary in the network with the help of autonomous system No. Less routing Table exchange is required.
4	It calculates the metric In terms of Hop Count from source network to destination network.	It calculates the metric In terms of bandwidth and delay (default).
5	RIP works on Bellman Ford algorithm.	EIGRP works on DUAL(Diffusing Update Algorithm) Algorithm.
6	It only maintains the best route to each destination.	It maintains the best route and some other alternative routes for each destination.
7	It is basically use for smaller size organization.	It is basically use for medium to lager size organization in the network.

7.1 Metric Calculation

EIGRP calculates metric based on four different parameters: bandwidth, delay, reliability, and load. While the first two are constant for a specific route, the rest are time dependent and are measured by routers. The default metric considers only the first two parameters: bandwidth and delay.

Table III Default bandwidth and delay

Interface	Bandwdith	Delay (microseconds)
Serial (T1)	1544 Kbps	20,000
Ethernet	10 Mbps	1000
Fast Ethernet	100 Mbps	100
Gigbit Ethernet	1000 Mbps	10
10 Gigbit Ethernet	10 Gbps	10

Bandwidth

Bandwidth is the lowest configured bandwidth of any interface along the route. The bandwidth of the exit interface of each router is considered. In Fig. 7.1, we see there are two routes from router 1 to network A: 1-4-2-A and 1-3-2-A. In the figure, bandwidth and delay are indicated by b and d, respectively. The bandwidths of 1 to 4, 4 to 2 and 2 to A are 56 kbps, 10,000 kbps and 10,000 kbps, respectively. The lowest bandwidth among these is 56 kbps; hence the bandwidth of the route 1-4-2-A is 56 kbps. Similarly, the bandwidth of another route is 128 kbps.

There is a default bandwidth of each interface depending on the interface type. Table III shows the default bandwidth of different types of interfaces. However, bandwidth of a network can be configured by using the following command in the global configuration mode:

```
Router(config-if)#bandwidth  <1-10,000,000>
```

Any value in the mentioned range can be selected. For example, if we want to set the bandwidth of the s0/0/0 serial interface to 56 kbps, we have to first select the interface; then, the bandwidth command can be used. For example,

```
Router(config)#interface s0/0/0

Router(config-if)#bandwidth  56
```

It would be pertinent to mention here that this `bandwidth` command has no effect on the actual data rate of the interface. The actual data rate depends on the clock rate at DCE (data communication equipment) end of the link. The purpose of the `bandwidth` command is to influence the routing protocol's decision in deciding a route.

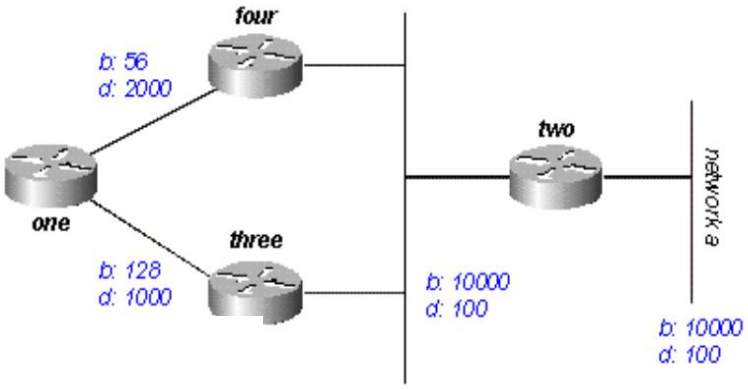

Fig. 7.1 A sample network I

Delay

Delay is calculated as the sum of delays from source to destination in units of 10 microseconds. Let's consider Fig. 7.1 again. The delays for 1 to 4, 4 to 2 and 2 to A are 2000, 100 and 100, respectively. The total delay is 2000 + 200 + 200 = 2200. Since the unit of day is tens of microsecond, the total delay of 1-4-2-A is 2200/10 = 220.

Similar to the bandwidth, the delay depends on the interface type. The default delay for different interfaces are given in Table III. However, we can configure bandwidth of an interface using the following command in global configuration mode:

```
Router(config-if)#delay < 10 to 167,772,140 >
```

Any value in the mentioned range can be selected. For example, if we want to set the delay of the s0/0/0 serial interface to 100 microseconds, we have to first select the interface; then, then the delay command can be used. For example,

```
Router(config)#interface s0/0/0
Router(config-if)#delay  10
```

How to calculate metric of a route?

To calculate the default metric of a route, we have to use the following formula:

$$Metric = \left[\frac{10^7}{least\ bandwidth} + delay_{total}\right] \times 256$$

Let's calculate the metric of the 1-4-2-A route of the network given in Fig. 7.1. We have already computed the bandwidth of the network: 56 kbps and delay: 220 microseconds. Putting these in the above equation, we get,

$$Metric = \left[\frac{10^7}{56} + 220\right] \times 256 = 45770496$$

Similarly, we can compute the metric for the 1-3-2-A route.

7.2 Neighbor Discovery

When a router connected to a network is turned on, its first task is to know its neighbors, where the neighbors are those routers which are directly connected to the router and satisfy certain criteria. A router follows a procedure to discover its router. The procedure is illustrated in Fig.7.2. The procedure starts by the new router, R1, with sending a multicast packet named *Hello* through all of its active interfaces. The multicast IP address is 224.0.010. The *Hello* packet contains the autonomous systems number of the autonomous system to which the sender belongs. In addition, it contains k-values: k_1, k_2, k_3, k_4, and k_5 which are used for calculating measured metric of a route. For sake of simplicity, we are not going into the details of measured metric calculation. The packet also consists of other parameters such as authentication information (optional) and so on. After receiving the *Hello* packet, R2 matches the autonomous system number and k-values of R1 with its own values. If all parameters match, R2 becomes a neighbor of R1. It also sends a *Hello* packet to R1 along with its whole routing table. This time, R2 sends the packets to the IP address it recovered from the received *Hello* packet's source address. After getting the *Hello* packet from R2, R1 sends an acknowledgement to R2. Based on the received routing tables from its neighbors, R1 populates its routing table. Then, it sends the whole routing table to its neighbor R2, which is then acknowledged by R2.

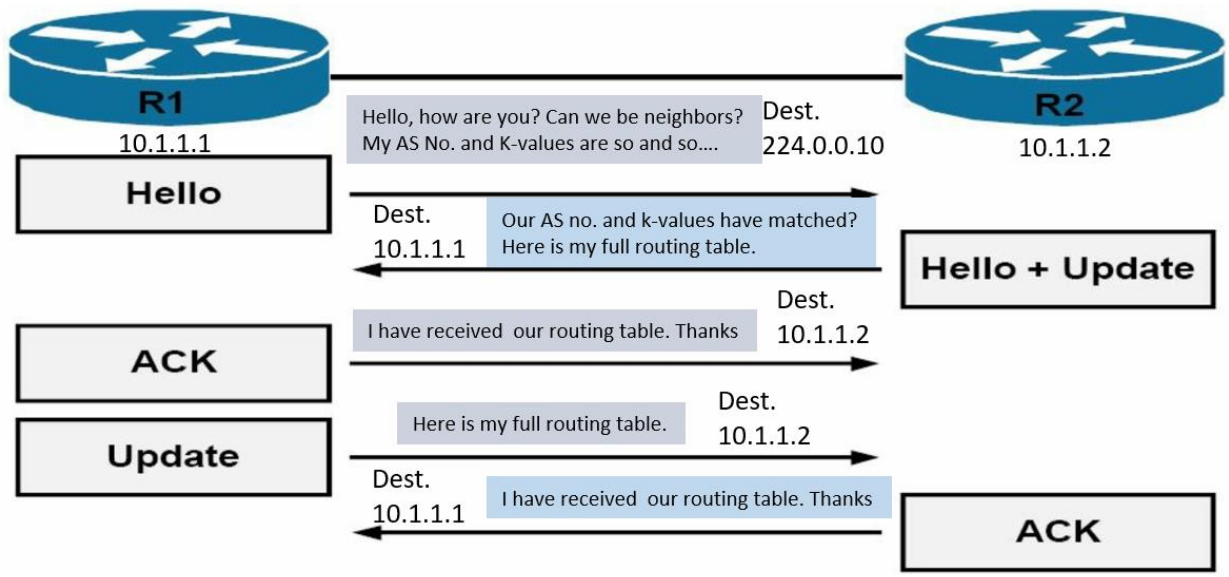

Fig. 7.2 Neighbor discovery

After establishing the neighborship, R1 and R2 sends a *Hello* packet to each other every 5 seconds (*Hello* time) to let each other know that they are still in Up state. If a router does not receive any *Hello* packet from its any neighbor within 15 seconds (Hold time), it considers the neighbor as Down and removes the routes it learned from the neighbor from its topology (will be discussed later) and routing table. It also sends such change to its other neighbors. However, the *Hello* and hold times are 60 seconds and 180 seconds for low-bandwidth networks such as serial leased line (T1).

Reader should keep in mind that a router with EIGRP sends its whole routing table to its neighbor during the neighbor discovery stage. In any other cases, it sends only that portion of the table that has new information. This reduces the bandwidth consumption compared to the sending of whole table as is done in RIP. In addition, instead of sending update at regular interval as is done by RIP, EIGRP only sends any update when there is any change in its routing table. Such infrequent update reduces bandwidth consumption, thereby ensuring higher throughput.

7.3 EIGRP Tables

RIP maintains only one table called routing table which stores information about the best route to each destination network. However, EIGRP maintains three different tables. Before describing each table, let's first get introduced with some terms.

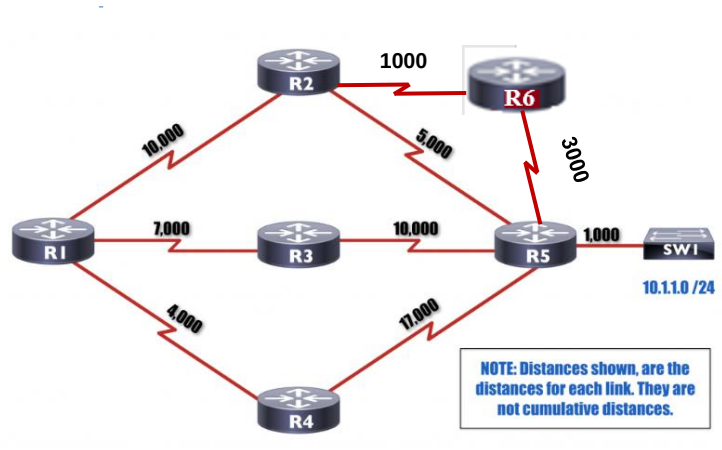

Fig. 7.3 A sample network II

Fig. 7.3 shows a sample network. Consider the R1 router. It has three neighbors: R2, R3, and R4. It gets information from each of its neighbor about any destination networks A, say 10.1.1.0/24. The neighbor R2 has two routes to the destination: R2-R5-A and R2 R6-R5-A. The metric for the first route is 5000+1000 = 6000, while it is for the second route is 1000+3000+1000 = 5000. Since the second route has lower metric, it will send (also refers to report or advertise) the metric 5000 to R1. For the router R3, it has only one route to A: R3-R5-A. It reports 11,000 (=10,000+1000) to R1. Similarly, R4 reports 18000 (=17,000+1000) to R1. The metric sent from each neighbor is called **reported distance** from the respective neighbor.

After getting reported distance, F1 adds its distance to each neighbor with the reported distance received from the respective neighbor. For example, R1 adds 10,000 to the reported distance 5000 reported from R2 to get 10,000+5000 = 15,000. This summation is called the **feasible distance** from R1 to A via neighbor R2. Similarly, the feasible distance from R1 to A via R3 and R4 are 7000+11,000 = 18,000 and 4000+18,000 = 22,000, respectively.

After getting all the feasible distance, R1 selects the best route to A. The route having the lowest feasible distance is called successor. Since the routes via R2, R3 and R4 have feasible distances of 15,000, 18,000 and 22,000, respectively, the feasible distance via R2 is the lowest. Hence, the best route to A from R1 is R1-R2-R6-R5-A. The best route is called the **successor**.

EIGRP uses diffusing update algorithm (DUAL) to ensure a loop-free successor. DUAL considers a route loop-free if the reported distance from a neighbor is less than the feasible distance of the successor. For the given network, the feasible distance of the successor (R1-R2-R6-R5-A) is 15000. Those routes which have reported distance less than the feasible distance, 15,000, of the successor are called feasible successors. The route via R3 has reported distance 11,000 which is less than the feasible distance (15,000) of the successor, hence the route via R3 is a feasible successor from R1 to A. On the other hand, the route via R4 has reported distance of 18,000 which is greater than the feasible distance (15,000) of the successor, hence the route via R3 is not a feasible successor for the route from R1 to A.

EIGRP maintains a table, named **topology table**, that stores information about the successor and the feasible successors for each destination network. EIGRP allows to store up to 32 feasible

successor. The feasible routes are stored as backup routes in the topology table. The successor is copied to another table, called **routing table**. The routing decision is made based on this table content. Thus, the routing table contains the shortest path for each destination network. If any route fails, EIGRP selects the best feasible route from the topology and copies the route to the routing table as the successor.

EIGRP maintains another table, named **neighbor table**. The routing table stores information about each neighbor. For each neighbor, it stores the IP address of the neighbor's interface (which is connected the router) and its own interface name which the neighbor is connected to.

Let's consider the network shown in Fig. 7.4 to illustrate conceptual tables of EIGRP. We have three routers: Corp, SF and NY. We are now talking about the contents of the Corp table. Suppose that the destination network is 10.0.0.0/8. Since Corp router is directly connected to two routers having the same EIGRP properties, the neighbor table contains SF and NY. Actually, instead of storing router names, neighbor table contains the IP address of the interfaces of SF and NY to which Corp is connected. It also contains it those interfaces names which the neighbors are connected to.

Suppose that the route to the destination via SFR has lower metric than the metric of the route via NY. Then, the route through s0/0 interface is successor. It is also assumed that the reported distance from NY is smaller than the feasible distance of the successor. Then, we can call the route through s0/1 interface of Corp is a feasible successor. The successor is then copied to the routing table. If the route through s0/0 (the successor) fails, the feasible successor from the topology table will be copied to the routing table. All routing is done using the successor. For each destination, there will be an entry in the routing table. However, there can be up to 32 feasible successors for each destination in the topology table.

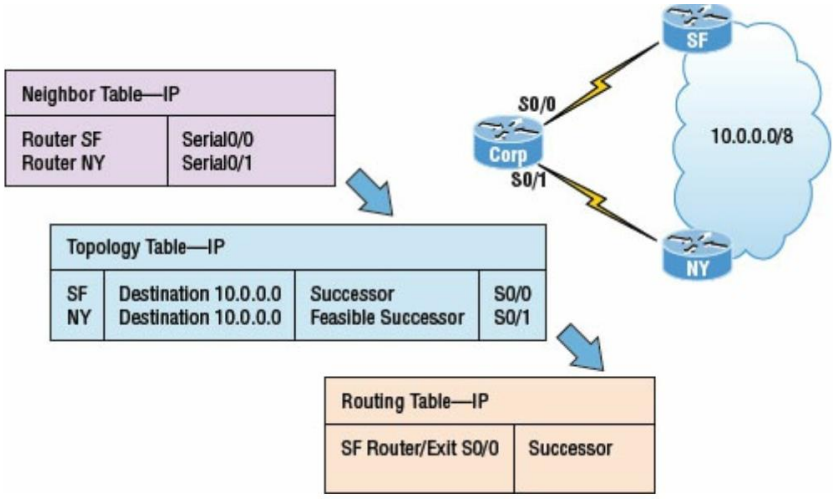

Fig. 4 Conceptual illustration of different EIGRP tables.

Exercise

1. Calculate metric for all possible routes from router ONE to network A in the following network.

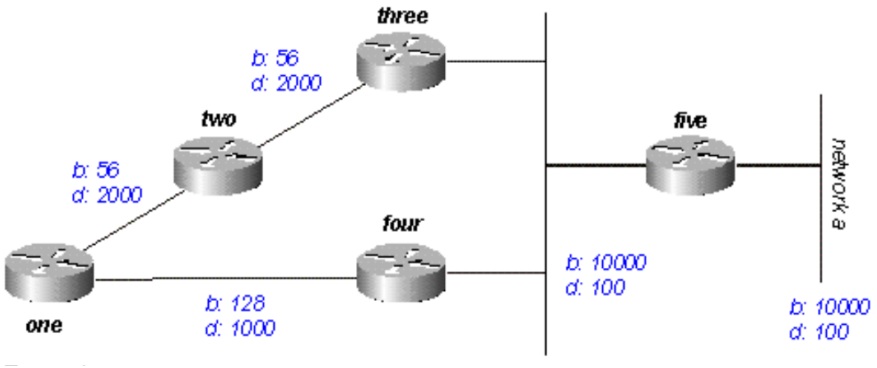

Fig. 7.5 Exercise 1

2. Configure the following network using EIGRP. Also verify the configuration using the verification commands.

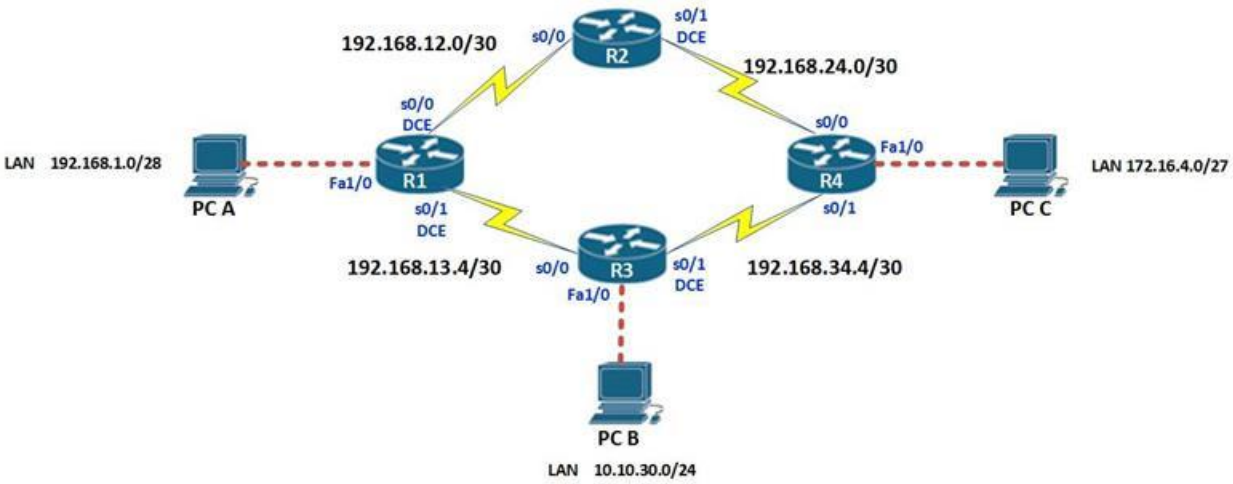

Fig. 7.6 Exercise 2